

Assignment: Decision Tree Classification

Group Size: 3 students

Deadline: Friday, 21 August 2026

Group Registration

One member from each group must fill out the following form and provide the **roll numbers** of all three team members by **18th August, 2026**:

<https://forms.gle/GrH9JquteUFRAWHHA>

Dataset

Use the **UCI Adult (Census Income) Dataset**.

The objective is to predict whether a person's annual income is $>50K$ or $\leq 50K$.

Dataset page:

<https://archive.ics.uci.edu/dataset/2/adult>

Direct download:

<https://archive.ics.uci.edu/static/public/2/adult.zip>

Use:

- `adult.data` for training and validation,
- `adult.test` for final evaluation,
- `adult.names` for attribute descriptions.

Split `adult.data` into **80% training and 20% validation**. Use a fixed random seed for the split.

Task 1: Implement a Decision Tree from Scratch

Implement a Decision Tree classifier **from scratch**.

Your implementation must support:

- categorical attributes,
- continuous/numerical attributes,
- missing values,
- multiway splits for categorical attributes,
- binary threshold splits for continuous attributes,
- deterministic tie-breaking.

Implement the following splitting criteria:

1. Gini Index
2. Entropy / Information Gain

3. Gain Ratio
4. Misclassification Error

For continuous attributes, determine the best threshold by considering valid candidate split points from the sorted training values.

Existing implementations of Decision Trees or tree-induction algorithms **must not be used**.

Task 2: Investigate Split Selection

At the **root node**, independently determine the best split using:

- Gini Index,
- Information Gain,
- Gain Ratio,
- Misclassification Error.

Report the feature and split selected by each criterion.

Additionally:

1. Find at least **one node where two splitting criteria prefer different splits**. Show the relevant calculations and explain the disagreement.
2. Select one numerical feature and demonstrate how its optimal threshold was selected.
3. Select one categorical feature containing at most 8 distinct values and compare **multiway splitting with all valid binary partitions**.
4. Show the complete manual calculation for at least one non-trivial split.

Task 3: The Information Gain Trap

Add an artificial attribute to the training and validation data:

```
UniqueID = 1, 2, 3, ..., N
```

Investigate what happens when this attribute is available to trees trained using:

- Information Gain,
- Gain Ratio.

Explain:

- why UniqueID can appear to be an excellent feature according to Information Gain,
- why such a split is poor for generalization,
- how Gain Ratio behaves in comparison.

Also inspect the original Adult dataset and identify at least **one pair of features containing overlapping or redundant information**. Discuss how this may affect tree construction.

Task 4: Overfitting and Noise

Train trees using:

```
max_depth = 2, 4, 6, 8, 10, 15, unlimited
```

For each setting report:

- training accuracy,
- validation accuracy,
- number of nodes,

- number of leaves,
- actual maximum depth.

Plot training and validation performance against tree complexity.

Next, randomly flip **10% of the training labels** and repeat the experiment. The selected examples must be determined using a fixed random seed.

Task 5: Final Model

Based only on the training and validation sets, select your final model.

Evaluate the final model on `adult.test`.

Report:

- Accuracy,
- Precision,
- Recall,
- F1-score,
- Confusion Matrix,
- Number of nodes,
- Number of leaves,
- Maximum depth.

Justify how you selected the final model.

Allowed Libraries

The Decision Tree and all associated algorithms must be implemented by your group.

External machine-learning implementations or libraries must not be used.

You may use the following libraries:

- Python standard library,
- `numpy`,
- `pandas`,
- `matplotlib`,
- `seaborn`.
- `scikit-learn`, only for `train_test_split`

Reproducibility

Random seeds must be explicitly set everywhere randomness is involved.

This includes:

- train-validation splitting,
- label-noise experiments,
- random sampling,
- any other randomized procedure.

The submitted notebook must be fully reproducible.

It is expected that when the notebook is executed from beginning to end, the resulting scores, tables and figures will **match the values reported in the submitted report**.

Submission Format

Each group must submit **one ZIP file** containing exactly the following:

1. One **.ipynb** file containing the complete implementation and all experiments.
2. One **.pdf** report containing the methodology, calculations, results, figures and analysis and contributions of each of the members in the team.

There is **no page limit** for the report.

The ZIP file must be named using the roll numbers of all three group members:

For example:

2401CS01_2401CS17_2401CS42.zip

Submission of a .py file instead of the .ipynb notebook is NOT acceptable.

The notebook should execute from beginning to end and reproduce all results reported in the PDF.

Assignment Submission

Submit the final ZIP file using:

<https://forms.gle/chUhr11TAkcM4RpNA>

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